

Government Health Expenditure and Life Expectancy

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Abstract

Introduction

Relevant Literature

Research Question and Hypotheses

Dataset and Methods

Dataset

This study combines three international datasets to examine the relationship between government healthcare expenditure and life expectancy across countries between 2007 and 2017.

The main explanatory variable was obtained from the World Bank Data360 platform using the International Monetary Fund (IMF) Government Finance Statistics indicator `**IMF_COFOG_GEL_GF07` (Government Expenditure on Health)**, which measures general government healthcare expenditure as a percentage of GDP World Bank (2026b). The main outcome variable was obtained from the World Bank Data360 indicator `**WEF_GCIHH_LIFEEXPECT**`, which reports life expectancy measured in years World Bank (2026c). To account for differences in national economic development, GDP per capita was obtained from the World Bank World Development Indicators (WDI) using indicator `**WB_WDI_NY_GDP_PCAP_CD**` World Bank (2026a). A regional classification variable was additionally generated from ISO3 country codes using the ‘countrycode’ R package.

The unit of analysis is the country-year. After cleaning and standardizing the three datasets, they were merged using ISO3 country codes and year to construct the final analytical dataset. The final dataset contains 601 country-year observations across 64 countries covering the period from 2007 to 2017.

One limitation of the dataset is that the historical Government Health Expenditure indicator (2007–2017) is no longer publicly available through the current World Bank Data360 or IMF Government Finance Statistics interfaces. In addition, because the final analysis requires complete information across all model variables, only countries with complete observations were retained in the analytical sample.

Data Preparation

Data acquisition, cleaning, and merging were performed using reproducible R scripts within an R Project workflow. Life expectancy and GDP per capita data were downloaded directly from the World Bank Data360 platform using documented API requests and stored locally before analysis. The historical Government Health Expenditure dataset (IMF_COFOG_GEL_GF07) was originally downloaded during the project proposal stage. During preparation of the final project, attempts were made to reproduce the original download through both the current World Bank Data360 platform and the IMF Government Finance Statistics (GFS) Data Explorer. Because the historical 2007–2017 observations are no longer publicly accessible, the archived CSV retained from the proposal stage was used with instructor approval. The original data source, indicator information, retrieval process, and verification steps are documented in the project repository.

Each dataset was cleaned separately prior to merging. All datasets were restricted to the common study period of 2007–2017. For the health expenditure dataset, only general government expenditure observations measured as a percentage of GDP were retained. For the life expectancy dataset, observations reported in years were retained. GDP per capita observations were restricted to valid positive values, and the natural logarithm of GDP per capita was calculated because the original variable exhibited substantial right skewness. Country identifiers were standardized using ISO3 country codes, duplicate country-year observations were removed, and a regional classification variable was derived from ISO3 country codes using the ‘countrycode’ package.

The cleaned datasets were merged using country code and year. A complete-case analytical dataset was then created by retaining only observations with complete information for all variables included in the regression analysis. The resulting dataset contained 601 country-year observations across 64 countries and was used for all subsequent exploratory analyses and regression models.

Methods

Software

All data acquisition, cleaning, merging, exploratory analysis, and statistical modelling were conducted in R R Core Team (2024). Data manipulation and visualization were performed using the tidyverse collection of packages Wickham

et al. (2019), including readr Wickham et al. (2024), dplyr Wickham et al. (2026), and tibble Müller and Wickham (2023). Reproducible file paths were managed using here Müller (2025). Initial data inspection and cleaning used janitor Firke (2024). API-based data retrieval was implemented using httr2 Wickham (2025), regional classifications were generated using countrycode Arel-Bundock et al. (2018), and regression model outputs were summarized using broom Robinson et al. (2026).

Results and Findings

Discussion and Conclusion

References

- Arel-Bundock, Vincent, Nils Enevoldsen, and CJ Yetman. 2018. “Countrycode: An r Package to Convert Country Names and Country Codes.” *Journal of Open Source Software* 3 (28): 848. <https://doi.org/10.21105/joss.00848>.
- Firke, Sam. 2024. *Janitor: Simple Tools for Examining and Cleaning Dirty Data*. <https://CRAN.R-project.org/package=janitor>.
- Müller, Kirill. 2025. *Here: A Simpler Way to Find Your Files*. <https://CRAN.R-project.org/package=here>.
- Müller, Kirill, and Hadley Wickham. 2023. *Tibble: Simple Data Frames*. <https://CRAN.R-project.org/package=tibble>.
- R Core Team. 2024. *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing. <https://www.R-project.org/>.
- Robinson, David, Alex Hayes, Simon Couch, and Emil Hvitfeldt. 2026. *Broom: Convert Statistical Objects into Tidy Tibbles*. <https://CRAN.R-project.org/package=broom>.
- Wickham, Hadley. 2025. *Httr2: Perform HTTP Requests and Process the Responses*. <https://CRAN.R-project.org/package=httr2>.
- Wickham, Hadley, Mara Averick, Jennifer Bryan, et al. 2019. “Welcome to the tidyverse.” *Journal of Open Source Software* 4 (43): 1686. <https://doi.org/10.21105/joss.01686>.
- Wickham, Hadley, Romain François, Lionel Henry, Kirill Müller, and Davis Vaughan. 2026. *Dplyr: A Grammar of Data Manipulation*. <https://CRAN.R-project.org/package=dplyr>.

Wickham, Hadley, Jim Hester, and Jennifer Bryan. 2024. *Readr: Read Rectangular Text Data*. <https://CRAN.R-project.org/package=readr>.

World Bank. 2026a. *GDP Per Capita (Current US\$) (WB_WDI_NY_GDP_PCAP_CD)*. https://data360.worldbank.org/en/indicator/WB_WDI_NY_GDP_PCAP_CD.

World Bank. 2026b. *Government Expenditure on Health (IMF_COFOG_GEL_GF07)*. https://data360.worldbank.org/en/indicator/IMF_COFOG_GEL_GF07.

World Bank. 2026c. *Life Expectancy, Years (WEF_GCIHH_LIFEEXPECT)*. https://data360.worldbank.org/en/indicator/WEF_GCIHH_LIFEEXPECT.